Sheet: VeraModule

VERA FPGA

File: vera-fpga.sch

Sheet: BusDecoder

BUS DECODER

File: busdecoder.sch

Sheet: Vera FPGA flash

VERA SPI FLASH
SD CARD INTERFACE

File: vera-fpga-flash.sch

Sheet: CartridgeInterface

CARTRIDGE INTERFACE

File: cartridgeInterface.sch

Sheet: PowerSupply

POWER SUPPLY

File: powersupply.sch

Sheet: VGA Analog

VGA ANALOG

File: vga-analog.sch

FIDUCIAL TOP

FID3 Fiducial

FID4 Fiducial

FIDUCIAL BOTTOM

FID2 Fiducial

FID1 Fiducial

LOGO1
VERA X16 LOGO

LOGO2
ATARI READY

LOGO3
KiCAD DESIGN

LOGO4
ATARI DUO BUS

LOGO5
ESP32_BOBOARD

LBL1

H1 MountingHole

H2 MountingHole

H3 MountingHole

H4 MountingHole

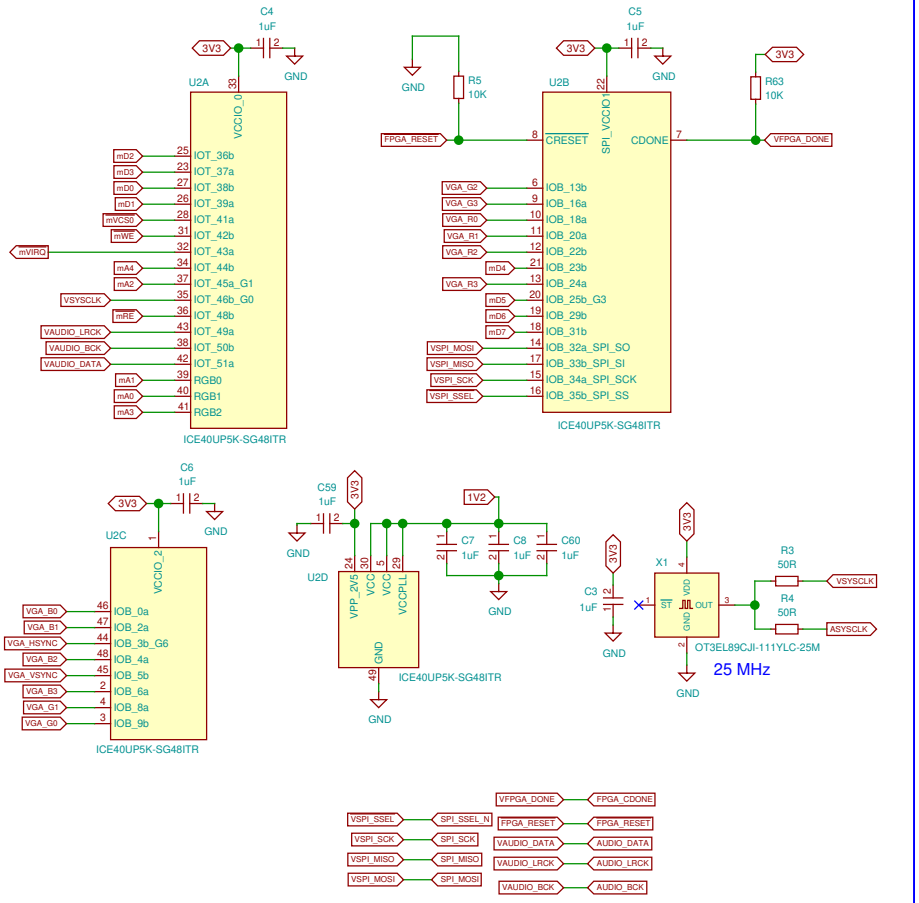
Gianluca Renzi
RetroBit Lab

Sheet: /
File: VERA-MODULE-RBL.sch

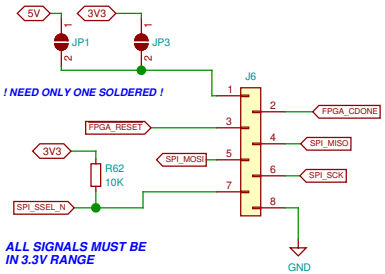
Title: VERA FPGA Audio & Video Board

Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261102-09:44Z		Id: 1/7

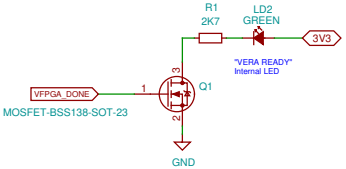
FPGA VERA LOGIC VIDEO & AUDIO CARD



iceprog programmer USB FTDI / SPI

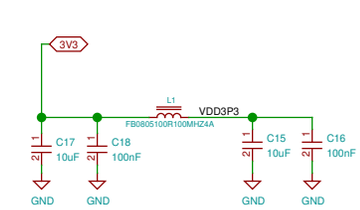


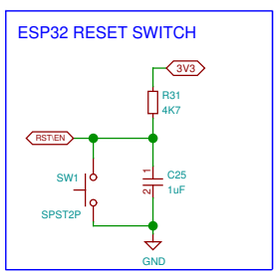
VERA FPGA PROGRAMMED OK

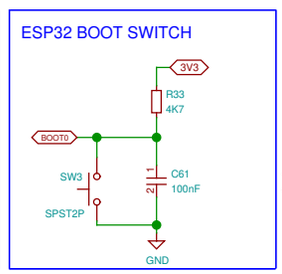


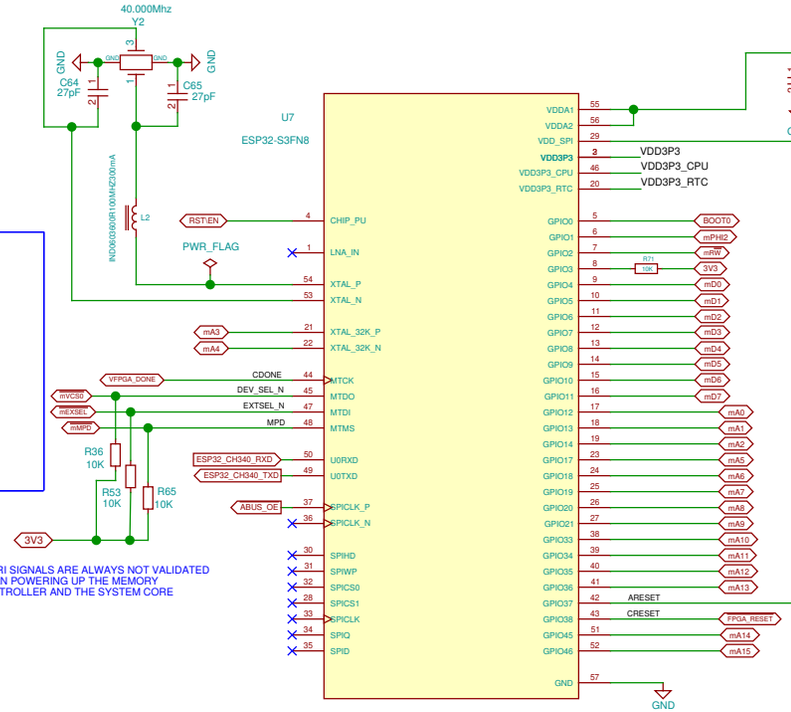
Gianluca Renzi		
RetroBit Lab		
Sheet: /VeraModule/		
File: vera-fpga.sch		
Title: VeraModule		
Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261102-09:44Z		Id: 2/7

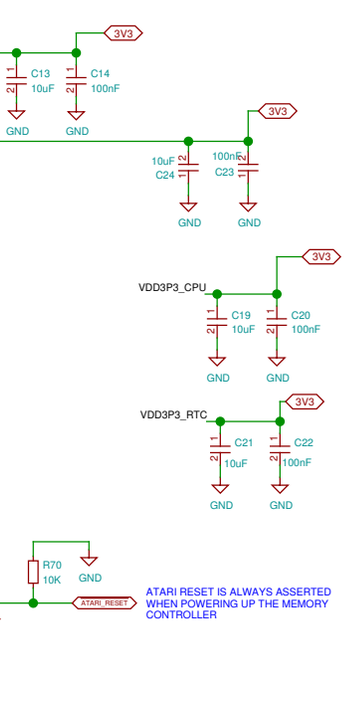
PBI DEVICE ID: software selectable only

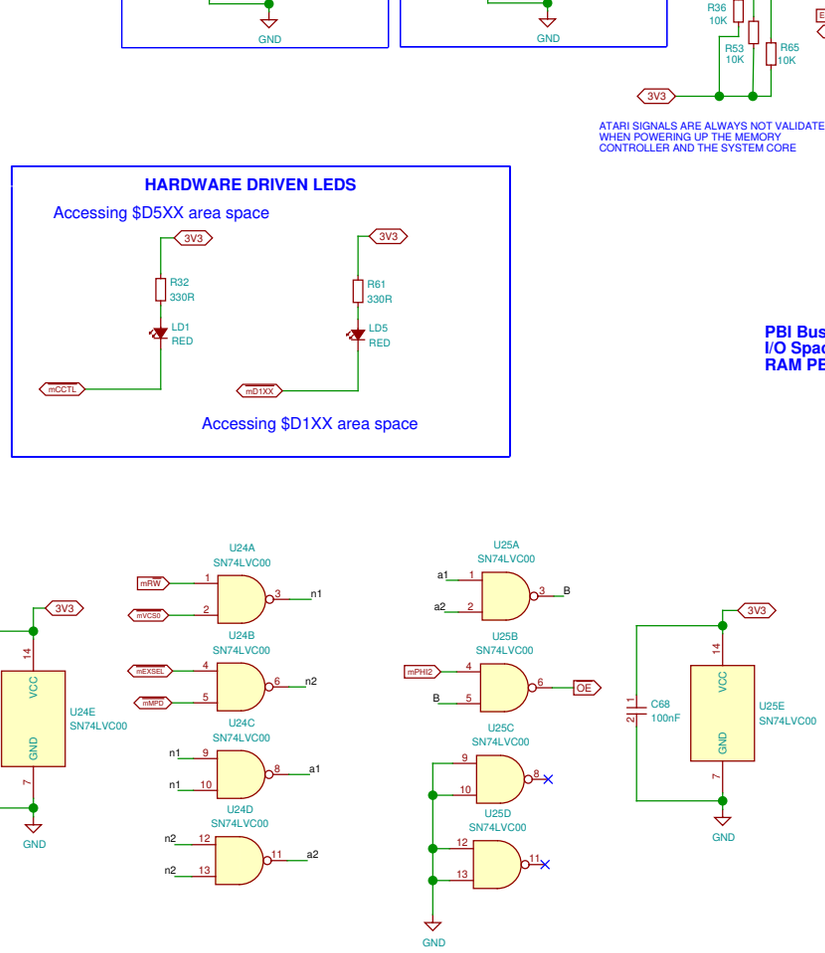




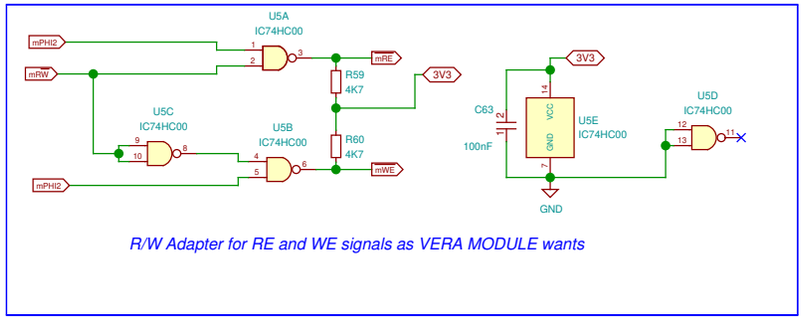








PBI Bus Interface Decoder:
I/O Space, Math Pack Disable, ROM ADDRESS ($D800-$DFFF),
RAM PBI SPACE ($D600-$D7FF)



When VERA Chip Select (mVCS0) latch is active & A15..A0 $D800-$DFFF -> MPD active (External 2K ROM)
When VERA Chip Select (mVCS0) latch is active & A15..A0 $D600-$D7FF -> EXSEL active (External 512 bytes RAM)
When access @ $D1xx and VERA Chip Select (mVCS0) latch is active -> EXSEL active

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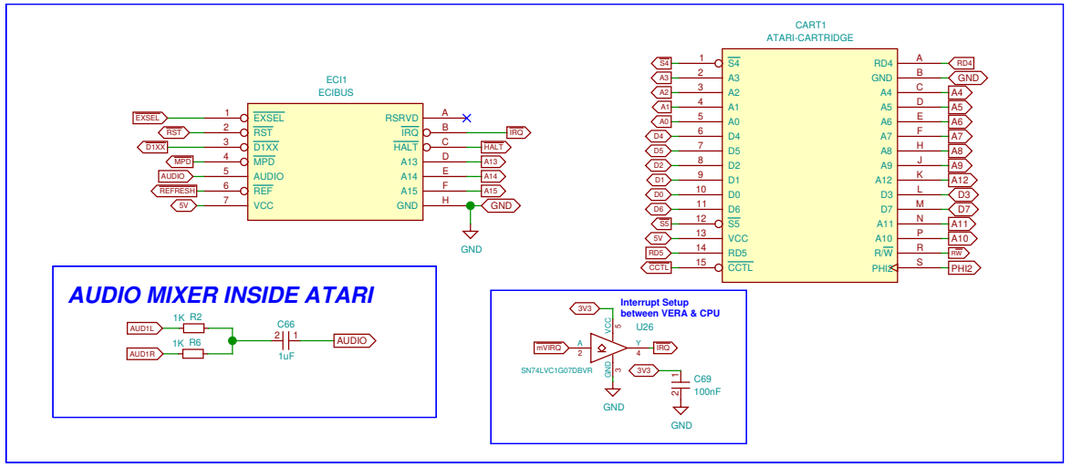
Sheet: /BusDecoder/
File: busdecoder.sch

Title: BusDecoder

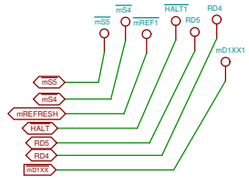
Size: A3	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261102-09:44Z		Id: 3/7

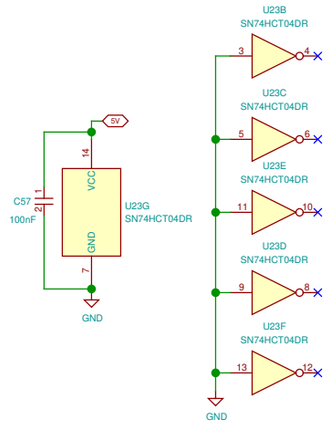
ATARI 130XE ECI & CARTRIDGE INTERFACE

All 8-Bit signals must be shifted from 5V to 3.3V and vice versa

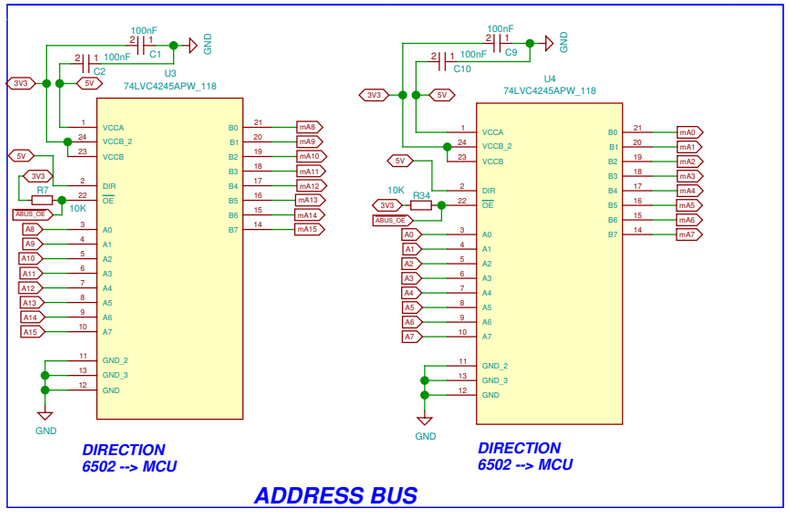


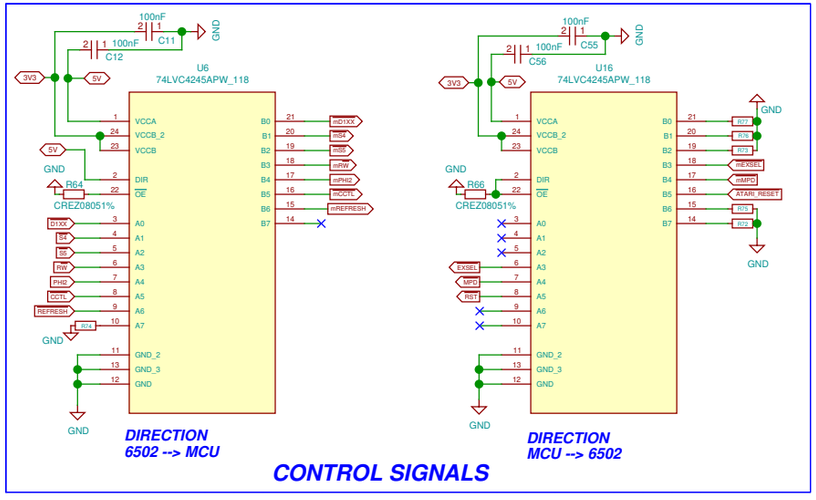
UNCONNECTED IN THIS PROJECT

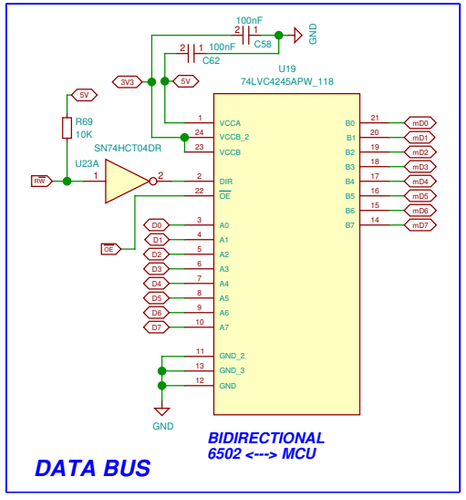




BUS LOGIC LEVEL SHIFTERS







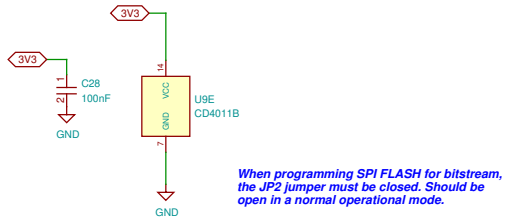
Gianluca Renzi
RetroBit Lab

Sheet: /CartridgeInterface/
File: cartridgeInterface.sch

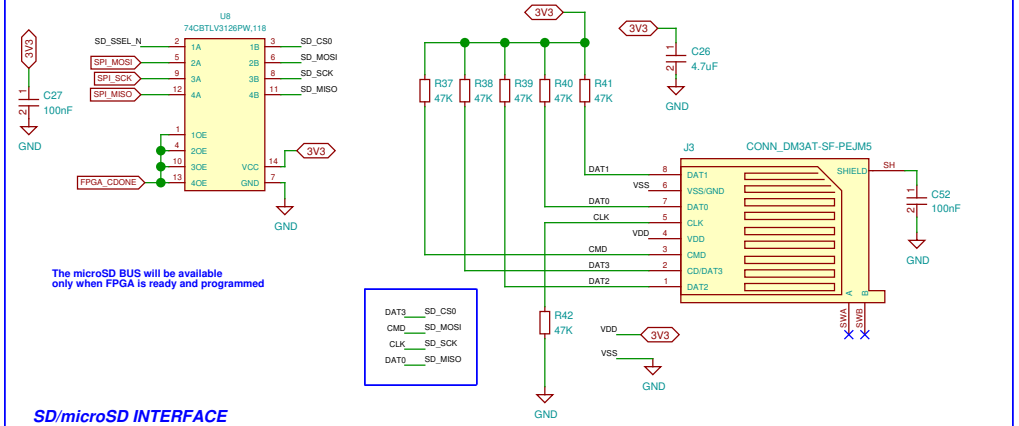
Title: CartridgeInterface

Size: A3	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261102-09:44Z		Id: 4/7

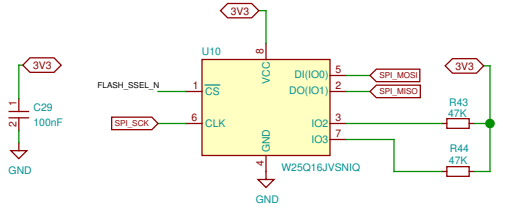
FPGA/SSD Flash Glue Logic



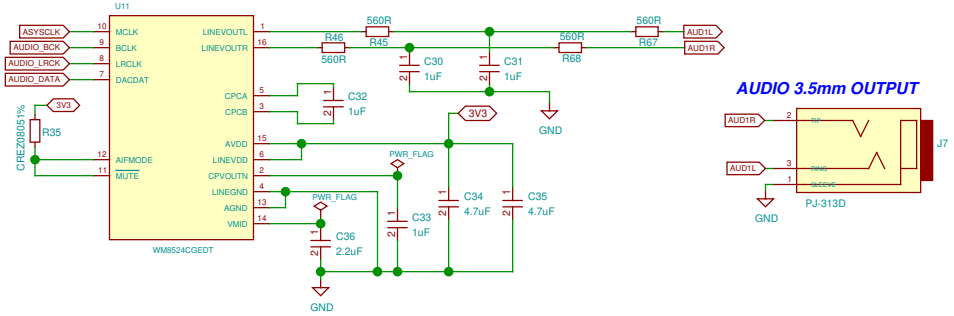
SD/microSD INTERFACE



SPI 16MB FLASH



IC DAC/AUDIO 24BIT 192K 16TSSOP



AUDIO 3.5mm OUTPUT

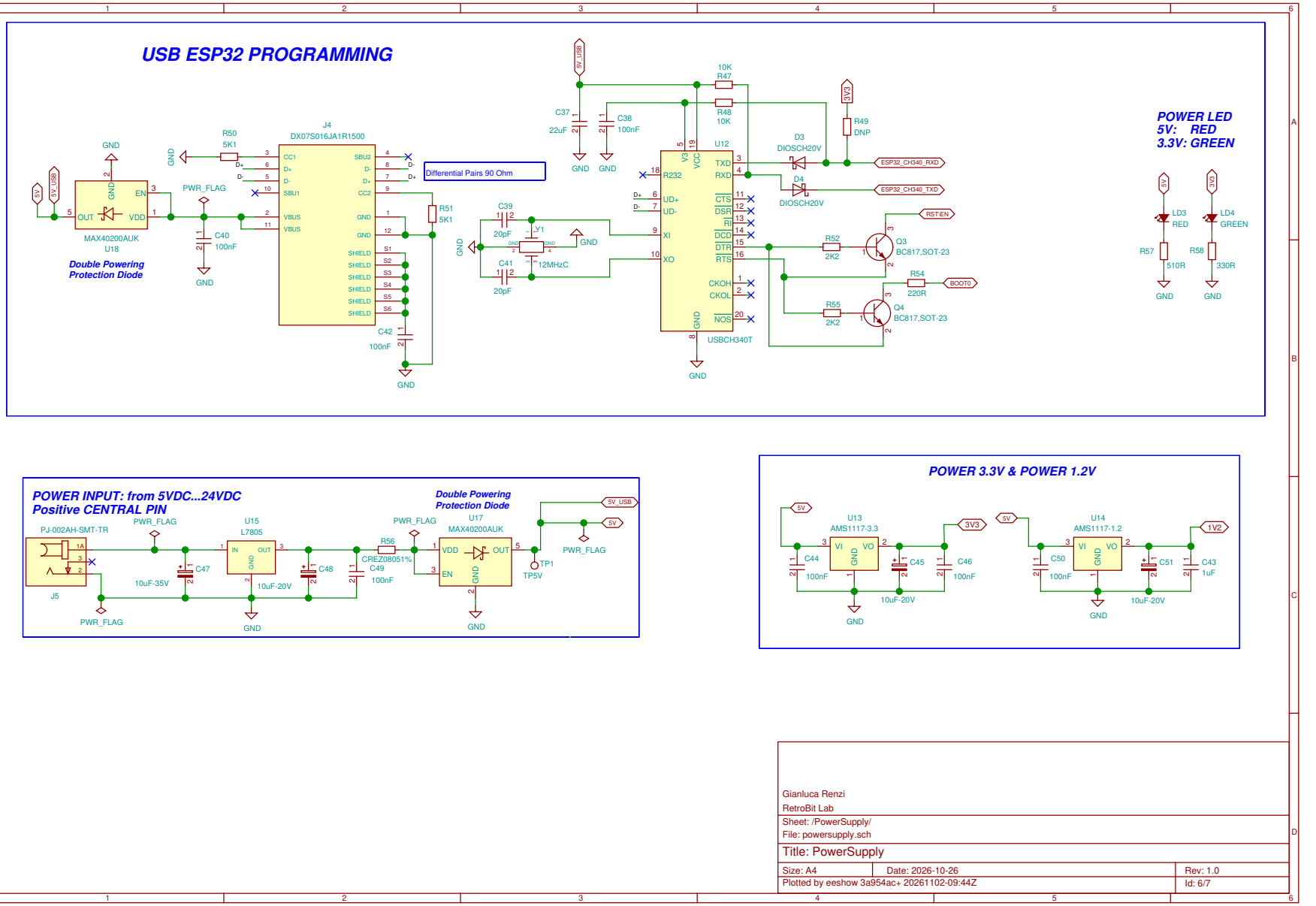
AUDIO SECTION

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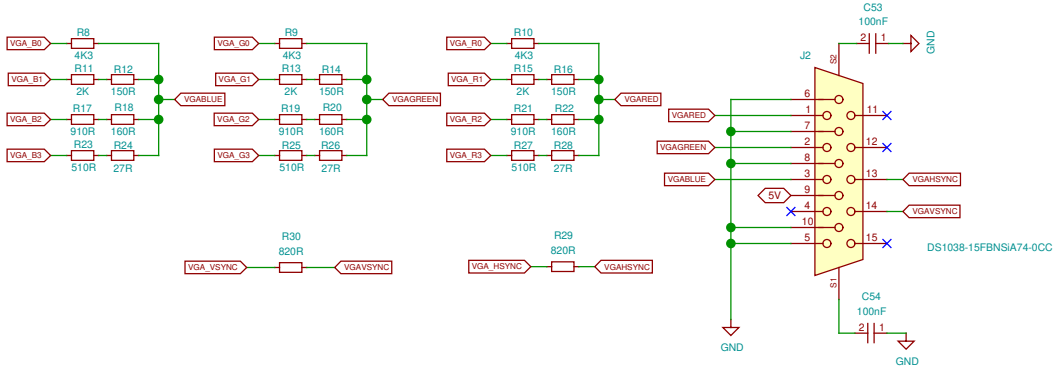
Sheet: /Vera FPGA flash/
File: vera-fpga-flash.sch

Title: Vera FPGA flash

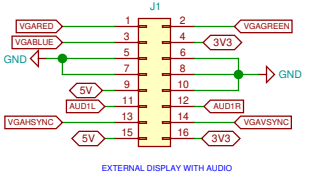
Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261102-09:44Z		Id: 5/7


USB ESP32 PROGRAMMING
POWER LED
5V: RED
3.3V: GREEN
Differential Pairs 90 Ohm
Double Powering
Protection Diode
POWER INPUT: from 5VDC...24VDC
Positive CENTRAL PIN
POWER 3.3V & POWER 1.2V
MAX40200AUK
U18
J4
DX07S016JA1R1500
U12
USBCH340T
U15
L7805
U17
U13
AMS1117-3.3
U14
AMS1117-1.2
Q3
BC817,SOT-23
Q4
BC817,SOT-23
D3
DIOSCH20V
D4
DIOSCH20V
LD3
RED
LD4
GREEN
12MHzC
PJ-002AH-SMT-TR
J5
TP1
TP5V
ESP32_CH340_RXD
ESP32_CH340_TXD
RST'EN
BOOT0
Gianluca Renzi
RetroBit Lab
Sheet: /PowerSupply/
File: powersupply.sch
Title: PowerSupply
Size: A4
Date: 2026-10-26
Rev: 1.0
Plotted by eeshow 3a954ac+ 20261102-09:44Z
Id: 6/7

ANALOG VGA SIGNALS



EXTERNAL VIDEO CONNECTOR



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RetroBit Lab		
Sheet: /VGA Analog/		
File: vga-analog.sch		
Title: VGA Analog		
Size: A4	Date: 2026-10-26	Rev: 1.0
Plotted by eeshow 3a954ac+ 20261102-09:44Z		Id: 7/7